

StressID Research Pipeline - Q1 Target (IEEE JBHI)

Temporal Cross-Modal Stress Detection with Missing-Modality Robustness

Research Pipeline & Objectives (StressID) — Q1 Target: IEEE JBHI

Working title: Temporal Cross-Modal Stress Detection Under Missing Modalities: A Leakage-Free, Reproducible Benchmark on StressID

Status: Planning → Implementation **Primary target:** IEEE Journal of Biomedical and Health Informatics (Q1, IF 7.7) **Backup target:** Computers in Biology and Medicine (Q1, IF 7.0) **Stretch target (2027 follow-up):** IEEE Transactions on Affective Computing (Q1, IF 9.8)

1. Research Objectives

Primary Objective

Develop and rigorously validate a multimodal stress detection model that captures **temporal dynamics** of stress (buildup/decay within a task), fuses **physiological, video, and audio** signals through **cross-modal attention**, and remains robust to **missing modalities at inference time** — evaluated under a **leakage-free (subject-grouped) protocol** that corrects inflated baselines in prior StressID work.

Secondary Objectives

- O1 — Corrected Benchmark:** Re-establish reliable baseline numbers for StressID using strict GroupKFold (no subject leakage across train/test), and quantify how much prior reported performance (yours and others') was inflated.
- O2 — Temporal Modeling:** Show that windowed, sequential modeling (30s windows → LSTM/Transformer) outperforms whole-task static feature aggregation, and measure the effect size.
- O3 — Cross-Modal Fusion:** Show that attention-based fusion (each modality attends to the others) outperforms concatenation/decision-fusion baselines, with an ablation isolating the fusion mechanism's contribution.
- O4 — Missing-Modality Robustness:** Show the model degrades gracefully (not catastrophically) when one or two modalities are absent at inference, using masked-token training (modality dropout).
- O5 — Cross-Dataset Generalization:** Test StressID-trained model on WESAD (ECG/EDA subset) without retraining, to support a generalization claim.
- O6 (stretch) — Class Imbalance / Minority Recall:** Improve 3-class affect recall on the minority classes (class-0, class-1) relative to baselines, since current methods collapse toward class-2.
- O7 — Data Scarcity via Multi-Dataset Pretraining:** Address StressID's small size (370 samples) by pretraining shared encoders across multiple related multimodal stress/emotion datasets before fine-tuning on StressID, and quantify the gain via a data-scaling ablation.

Auxiliary Datasets (for O5 / O7)

Dataset	Subjects	Overlapping modalities	Role in pipeline
WESAD	15	ECG, EDA, respiration	Physio-encoder pretraining + zero-shot transfer target (E7)
SWELL-KW	25	ECG/EDA, video/posture	Naturalistic-stress pretraining source; no audio (tests missing-modality masking)
K-EmoCon	32	EEG, ECG, BVP, EDA, audio, video	Broadest modality overlap; primary multi-dataset pretraining source
MuSE	28	Physio, audio, video	Closest in design intent (TSST-style induction) to StressID
ForDigitStress	40	Physio, audio, video	Similar size/protocol class; direct pooling candidate
BESST	79	Physio (+ audio/video varies)	Largest cohort; biggest sample-size boost
EmoWork	—	Physio, audio, video	Newest (2025/2026), workplace scenario; low-competition novelty angle

Explicit Novelty Positioning

Three closely related works now exist on StressID: H-C3AT-G (cross-modal attention + graph fusion, Feb 2026), an XAI/SHAP explainability study (May 2026), and a masked-multimodal-transformer paper that explicitly handles missing modalities on StressID using contrastive video-alignment. Because of this, “cross-modal attention” and “missing modality” are each **individually claimed already**. This paper's novelty must sit at the **intersection nobody has tested**: does temporal/windowed modeling change how gracefully a model degrades when a modality is missing? None of the three existing papers use windowed/sequential inputs — they all operate on one static feature vector per task. This paper differentiates itself by: - Being the first to combine **temporal windowing** with **missing-modality robustness** on StressID, and directly testing whether temporal context helps the model *compensate* for a missing modality better than a static model can (this is the paper's central research question, not a side ablation). - Delivering a **leakage-corrected benchmark** (GroupKFold) as a standalone, citable contribution — none of the three related papers report a GroupKFold-corrected baseline table. - Adding calibrated uncertainty on predictions made under missing-modality conditions — not reported anywhere in the existing StressID literature.

Revised central research question: “Does temporal context help a multimodal stress model degrade more gracefully under missing modalities, compared to a static (single-vector) model?”

Q1 Rigor Requirements (non-negotiable for JBHI/CBM submission)

1. **Re-implement and directly benchmark against all 3 competitor papers** (H-C3AT-G, StressID-XAI, StressID-missing-modality paper) in your own results table — citing them is not enough at Q1.
2. **Statistical significance on every headline result** — bootstrap 95% CI or paired t-test across ≥ 5 seeds/folds, not single-run numbers.
3. **Full public reproducibility package** — code, fixed GroupKFold splits, trained weights (GitHub, referenced in the paper).
4. **Calibration reporting** (ECE, reliability diagrams) wherever missing-modality predictions are shown — reviewers at health-informatics venues expect confidence to be honest, not just accuracy.
5. **At least one real-world/generalization angle** beyond in-dataset accuracy (your WESAD transfer, E7) — pure benchmark papers without any external validation are a common Q1 desk-reject reason.

2. Pipeline Overview

STAGE -1 – MULTI-DATASET PRETRAINING (addresses small-N problem)

Pool: WESAD + SWELL-KW + K-EmoCon + MuSE + ForDigitStress + BESST + EmoWork

- harmonize labels (map each dataset's stress/emotion labels to a common binary stress + 3-class affect schema)
- harmonize sampling rates (resample physio to 500Hz reference)
- each dataset contributes whichever modalities it has; ABSENT modalities → mask token (reuses Stage 3 missing-modality mechanism, so no separate alignment step needed)
- self-supervised / multi-task pretrain Stage 2 encoders + Stage 3 fusion block across the pooled corpus
- fine-tune full model on StressID (target domain)

Output: pretrained encoders + fusion block, ready for Stage 0

STAGE 0 – DATA & EVALUATION PROTOCOL FIX

StressID raw data (Physio 500Hz / Video / Audio)

- strict subject-level GroupKFold split (no leakage)
- re-run ALL existing baselines (unimodal, feature fusion, decision fusion)
- produce CORRECTED baseline table (this is Contribution #1)

STAGE 1 – TEMPORAL WINDOWING

Per subject-task recording (3-5 min)

- slide 30s windows (50% overlap) → ~8-10 windows/task
- label each window (task-level label propagated, or interpolated from continuous self-report if available)

STAGE 2 – MODALITY-SPECIFIC ENCODERS (per window)

Physio (ECG/EDA/RSP) → 1D-CNN + BiLSTM → Physio tokens [P1..Pn]
Audio (waveform) → wav2vec2 (fine-tuned) → Audio tokens [A1..Am]
Video (AU sequence) → per-frame Transformer → Video tokens [V1..Vk]

STAGE 3 – CROSS-MODAL FUSION (KEY CONTRIBUTION)

Multi-head cross-attention:
PhysioQ → AudioKV, VideoKV
AudioQ → PhysioKV, VideoKV
VideoQ → PhysioKV, AudioKV
+ CLS token aggregation
+ MODALITY DROPOUT during training (25% chance/modality masked)
→ learns to fuse gracefully with 1 or 2 modalities missing
→ Fused representation [512-d] per window

STAGE 4 – TEMPORAL AGGREGATION

Sequence of fused window representations → Transformer encoder (time axis)
+ task-type embedding conditioning
→ Task-level representation

STAGE 5 – OUTPUT HEADS

— Binary stress (sigmoid)
— 3-class affect (softmax)
— Stress score 0-10 (regression, optional/stretch)

Loss = BCE_binary + CE_3class + λ ·MSE_regression
+ contrastive subject-invariance loss (disentangle subject ID)

STAGE 5.5 – COMPETITOR RE-IMPLEMENTATION (Q1 requirement)

Re-implement (not just cite) the 3 direct competitors on your own GroupKFold splits, for apples-to-apples comparison:

- H-C3AT-G (cross-modal contrastive attention + graph fusion)
- StressID-XAI (SHAP-based explainability baseline)
- StressID missing-modality paper (contrastive video-alignment + masked multimodal transformer)

Output: a single results table with your model + all 3 competitors, all under the SAME leakage-free protocol (this table alone is a major reviewer-trust signal at Q1 venues)

1. Corrected baseline comparison (GroupKFold)
2. Ablation: static vs. temporal
3. Ablation: concat/decision fusion vs. cross-attention fusion
4. Robustness: full modalities vs. 1-missing vs. 2-missing (accuracy/F1 curve)
5. Cross-dataset: train StressID → test WESAD (ECG/EDA only)
6. Per-class recall breakdown (minority class focus)
7. Statistical significance (bootstrap CI or paired t-test vs. baselines)
8. Head-to-head vs. re-implemented competitors (Stage 5.5)
9. Calibration / ECE under missing-modality conditions

3. Experiment & Ablation Matrix

#	Experiment	Question it answers	Compares against
E0 (Q1-critical)	Re-implemented H-C3AT-G, StressID-XAI baseline, and missing-modality paper — all on your GroupKFold splits	How does your model directly compare, apples-to-apples, to the 3 real competitors?	Original competitor papers' reported numbers (sanity check) + your model
E1	Corrected baselines (GroupKFold)	How inflated were prior numbers?	Original StressID paper results
E2	Static features vs. temporal windows	Does temporal modeling help?	E1
E3	Concat/decision fusion vs. cross-attention	Does cross-modal attention help?	E1, E2
E4	Full model, full modalities	Best-case performance	All above
E5	Full model, 1 modality masked (×3 combos)	Graceful degradation?	E4
E6	Full model, 2 modalities masked (×3 combos)	Worst-case robustness	E4
E7	StressID→WESAD zero-shot transfer	Cross-dataset generalization	Published WESAD baselines
E8	Per-class recall (3-class)	Does the method fix minority-class collapse?	E1 baseline confusion matrix
E9 (stretch)	Frozen vs. fine-tuned wav2vec2	Audio encoder ablation delta	E4
E10	Data-scaling curve: StressID-only → +WESAD → +SWELL/K-EmoCon → +all sources	Does multi-dataset pretraining offset small-N?	E4 (StressID-only)
E11	Pretrained-and-fine-tuned vs. from-scratch encoders	Isolates the pretraining contribution specifically	E4, E10
E12 (HEADLINE)	Static model degradation curve (1/2 modalities missing) vs. Temporal model degradation curve (1/2 modalities missing)	Does temporal context make missing-modality degradation less steep? This is the paper's central claim.	E4, E5, E6
E13	Calibration / confidence under missing-modality conditions (reliability diagram, ECE)	Does the model know when it's less certain due to a missing modality?	E12

4. Success Criteria / What “Done” Looks Like

- ☐ Corrected GroupKFold baseline table matches or is clearly derived from original StressID protocol, with documented delta.
- ☐ Temporal model beats static-feature baseline by a statistically significant margin (report CI/p-value).
- ☐ Cross-attention fusion beats concat/decision fusion baseline, isolated via ablation.
- ☐ Missing-modality curve shows smooth degradation (not a cliff) — e.g., F1 drop < 10-15% with one modality missing.
- ☐ WESAD zero-shot transfer result reported, even if modest — demonstrates generalization was tested honestly.
- ☐ Minority-class (class-0/1) recall improves over baseline decision fusion.
- ☐ All results reproducible: code + fixed splits released (important for reviewer trust given the leakage narrative).
- ☐ Data-scaling curve (E10) shows a clear, monotonic (or near-monotonic) benefit from multi-dataset pretraining over StressID-only training.
- ☐ Re-implemented competitor comparison (E0) shows your model at least matching, ideally beating, all 3 competitors on the SAME leakage-free protocol.
- ☐ Calibration (E13) shows meaningfully worse confidence (higher ECE) under missing-modality conditions than full-modality — i.e., the model “knows” when it's uncertain.
- ☐ Public GitHub repo with code + splits + weights is live before submission (link goes in the paper).

5. Milestone Timeline (Q1-level rigor, ~18-20 weeks to submission)

Weeks	Milestone
1	Acquire/prepare auxiliary datasets (WESAD, SWELL-KW, K-EmoCon, MuSE, ForDigitStress, BESST, EmoWork); harmonize labels & sampling rates

2-3	Fix evaluation protocol; re-run & corrected StressID baseline table (O1)
4-5	Temporal windowing + physio-only temporal baseline (O2 partial)
6-7	Fine-tune wav2vec2; implement full modality encoders
8-9	Cross-modal attention fusion + modality dropout training (O3, O4)
10	Multi-dataset pretraining run (Stage -1) + fine-tune on StressID (O7)
11	WESAD cross-dataset test (O5) + data-scaling ablation (E10, E11)
12-13	Re-implement all 3 competitor papers on your GroupKFold splits (E0) — budget extra time, this is often harder than it sounds
14	Headline experiment: temporal vs. static missing-modality degradation curves (E12) + calibration (E13)
15	Full ablation matrix + statistical significance testing (≥ 5 seeds/folds) + minority-class analysis
16	Publish public GitHub repo (code, splits, weights); internal reproducibility check (re-run from scratch on a clean machine)
17-19	Draft writing, figures, related-work positioning, internal/advisor review
20	Final polish, formatting to JBHI author guidelines, submission

6. Paper Skeleton (for drafting, targeting JBHI format)

- Introduction** — stress detection matters for health monitoring; multimodal is necessary; current methods are static, leakage-prone, brittle to missing sensors, trained on small single-source datasets, and (critically) haven't been benchmarked against each other.
- Related Work** — StressID origin paper; H-C3AT-G (differentiate: no temporal, no missing-modality focus); StressID-XAI paper (differentiate: static SHAP vs. our temporal+robustness focus); StressID missing-modality paper (differentiate: no temporal windowing); WESAD/SWELL/K-EmoCon/other stress & emotion datasets; missing-modality transformer literature.
- Datasets** — StressID (target) + auxiliary pretraining pool (WESAD, SWELL-KW, K-EmoCon, MuSE, ForDigitStress, BESST, EmoWork); label/sampling-rate harmonization procedure.
- Method** — Stage -1 (multi-dataset pretraining) through Stage 5 of pipeline above.
- Evaluation Protocol** — leakage fix as a contribution in itself (E1); direct re-implementation protocol for competitors (E0).
- Experiments** — E0-E13 with tables/figures; E12 (temporal vs. static degradation curves) as the headline figure; E0 (head-to-head vs. competitors) as the credibility-anchoring table.
- Discussion** — clinical/practical implications of missing-modality robustness (wearable deployment angle); limitations (residual dataset heterogeneity, self-report label noise, StressID's remaining size even after pretraining).
- Reproducibility Statement** — GitHub link, exact splits, environment/seed details (JBHI and most Q1 venues now expect this as a dedicated statement).
- Conclusion & Future Work** — point toward your 2027 roadmap (naturalistic transfer, foundation-model backbone).

Generated as a working planning document — refine objectives and ablation matrix as implementation reveals what's actually feasible on your compute/time budget.